14 $g = \frac{Gm}{r^2}$

The value of r is decreased by 20%
The value of G and the value of m are not changed.

Calculate the percentage increase in the value of g.

New radius = $0.8r$

1 mark
Keep 80% of the radius.

New $r^2 = (0.8r)^2 = 0.64r^2$

g is multiplied by $1 \div 0.64 = 1.5625$

1 mark
Reciprocal of the squared factor.

Increase = $(1.5625 - 1) \times 100\%$

= 56.25%

1 mark
Subtract 1 then convert to%.

.....%

(Total for Question 14 is 3 marks)